

Building a Private Cloud Using OpenStack

OpenStack is a cloud computing platform that offers a suite of software tools for creating and maintaining both public and private clouds. It gives users the freedom to set up and control an unlimited number of virtual computers, networks, and storage devices. This short tutorial will show us how to set up a private cloud using OpenStack and launch a virtual machine over the deployed environment.



OpenStack architecture contains several components like controller nodes, compute nodes, neutrons, block storage, and object storage. Here's a brief description of these components (Figure 1).

Controller node: This node is responsible for providing most of the OpenStack services such as identity, image store, dashboard, etc.

Compute node (Nova): This is responsible for managing and automating compute instances (VMs) in the OpenStack environment.

Networking (Neutron): This enables networking connectivity and IP addresses for users.

Block storage node (Cinder): This node is used to provide persistent storage for the instances running at OpenStack.

Object storage (Swift): This is used for ensuring redundant, scalable, and fault-tolerant storage in the server cluster.

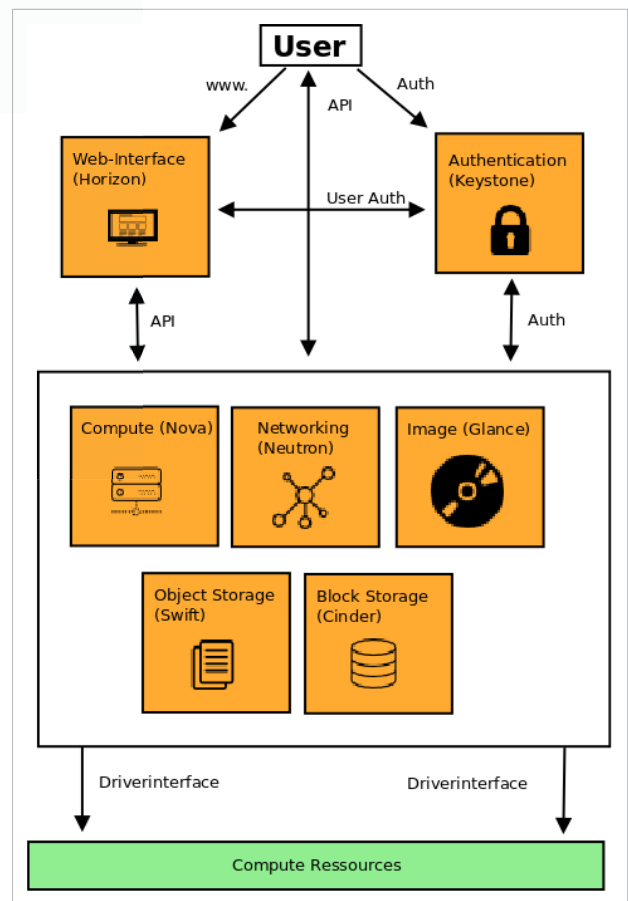


Figure 1: Key components in the OpenStack architecture